

Vasu Reddy Nalamalapu

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Experience

Brillio | Data Science Intern

Sept 2025 — Present
On-Site

- Assisted in automating internal business functions (Accounts Payable) using agentic AI solutions to reduce manual effort.
- Designed and collaborated on AI agents to orchestrate workflows, improve decision-making, and streamline enterprise processes.
- Developed a RAG agent to retrieve relevant SOPs, policies and historical records to assist agents in data validation and decision-making.
- Designed and implemented a RAG-based policy validation system to interpret and validate Corporate Credit Card spend policies, leveraging Python, Azure AI Search for vector retrieval, and Azure OpenAI for LLM-powered responses.
- Tech Stack:** LangGraph, Agentic AI, LLMs, Azure, PostgreSQL, Azure Data Studio, Python.

Projects

Agentic SQL Query Executor | Python, LangChain, ReAct Agent, MySQL, Streamlit

Jan 2026 — Feb 2026

- Built an **LLM-powered Agentic SQL Generator** that translates natural language questions into executable SQL queries, enabling non-technical users to query relational databases without writing SQL and reducing manual query dependency.
- Improved query accuracy and schema adherence by implementing a **ReAct-based LangChain agent** with dynamic schema injection and tool-based database execution, and deployed an interactive Streamlit interface for real-time validation and results visualization.

Multi-Document RAG Agent | Python, LangChain, RAG, FAISS, Groq API

Oct 2025 — Nov 2025

- Built a configurable **multi-document RAG agent** that ingests multiple PDFs, performs recursive chunking, and enables context grounded question answering using a Groq-hosted LLM.
- Designed a modular retrieval pipeline with **FAISS vector indexing**, dynamic embedding provider selection, similarity-based top-k retrieval, and source-aware response generation to ensure traceable and document-grounded outputs.

NYC Taxi Revenue Maximization | Python, SQL, Tableau, Scipy

Jun 2025 — Aug 2025

- Analyzed **10M+** NYC Yellow Taxi records to uncover revenue drivers, performing EDA, hypothesis testing, and predictive modeling that improved forecasting accuracy by **25%**.
- Built an interactive Tableau dashboard and geospatial visualizations (Folium) highlighting high-profit zones, enabling data-driven decision-making for fleet optimization.

Skills

- Programming & Query Languages :** Python, SQL
- AI skills :** LLMs, Prompt Engineering, ReAct agents, RAG, Embeddings
- Data Analysis & Statistical Modeling :** Data Cleaning & Preparation, Data Analytics, Statistical Analysis, Machine Learning
- Libraries & Frameworks :** NumPy, Pandas, SciPy, Matplotlib, Seaborn, Plotly, Dash, Folium, Scikit-learn, LangChain
- Databases & BI Tools :** MongoDB, MySQL, Tableau
- Tools & Platforms :** GitHub, Jupyter Notebook, Visual Studio Code
- Core Competencies :** Analytical Thinking, Attention to Detail, Problem Solving, Communication, Team Collaboration

Education

Vishnu Institute of Technology, BTech in Artificial Intelligence and Data Science

Oct 2022 — May 2026

- CGPA: 8.84*

Narayana Junior College, Intermediate (12th grade)

Jul 2020 — Apr 2022

- Percentage : 94.6%

Certifications & Achievements

- Contributed to **Smart India Hackathon** projects on Leaf disease Predictor.
- Earned **Elite + Silver Certificate** in NPTEL's "Data Analytics with Python" for excellence in Python programming and applied statistics.
- Certified in **IBM SQL for Data Science**, with emphasis on writing efficient queries, performing joins, and leveraging SQL for data analytics.